Shakti Prasanna Dash

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Professional Summary

Computer Science undergraduate at IIIT Bhubaneswar with hands-on experience building end-to-end Machine Learning and Computer Vision applications. Skilled in dataset engineering, deep learning, object detection, and deploying scalable AI systems using modern MLOps tools.

Technical Skills

Programming: Python, C, C++, SQL

ML/DL: PyTorch, TensorFlow, Scikit-learn, XGBoost, Keras, Optuna

Computer Vision: YOLO11, OpenCV

NLP: Hugging Face Transformers, LangChain, FAISS

Data Science: NumPy, Pandas, Matplotlib, Seaborn

Deployment & MLOps: Docker, Streamlit, FastAPI, Git, GitHub, MLflow, AWS

Database: MySQL

Education

Bachelor of Technology in Computer Science and Engineering *2023 – 2027*

IIIT Bhubaneswar

CGPA: **8.08**

Projects

Smart Traffic Violation Detection *Deep Learning • Computer Vision • MLOps*

[GitHub] *[Live Demo]*

Technologies: Python, YOLO11s, PyTorch, OpenCV, Streamlit, Docker, Optuna

- Built an end-to-end motorcycle helmet violation detection system using a custom-trained **YOLO11s** model with rider-helmet association, configurable helmet/turban compliance policies, and image, video, and webcam inference.
- Engineered a custom dataset of **3,035 images** and **12,254 annotations** by merging multiple public datasets (including EdgeVision), correcting annotation errors, creating a new **faceWithTurban** class, and augmenting minority classes.
- Performed dataset verification, class-distribution analysis, annotation validation, confusion-matrix evaluation, and Optuna-based hyperparameter tuning, achieving approximately **89% mAP@50**, **60% mAP@50–95**, **85% Precision**, and **84% Recall** on the validation set.
- Modularized the project into reusable detection, association, policy, visualization, and inference pipelines, then Dockerized and deployed it on Render using the CPU-only PyTorch build to satisfy free-tier memory constraints.

Laptop Price Predictor *Machine Learning • Regression*

[GitHub]

Technologies: Python, Scikit-learn, Pandas, Streamlit, XGBoost

- Built an end-to-end regression pipeline to predict laptop prices using hardware specifications collected from real-world e-commerce datasets.
- Performed exploratory data analysis, feature engineering, preprocessing, and model comparison across Linear Regression, Ridge Regression, Random Forest, and XGBoost.
- Achieved an RMSE below **3000** and deployed the final model through an interactive Streamlit web application.

Certifications

- CampusX Data Science Mentorship Program (2025 – Ongoing)
- Google Cloud Foundations – Cloud Jam GEN AI (2023)